Inhibiting LINE-1 retrotransposons therapeutically in disease

William Brandler[1]

[1]L1 Therapeutics, william@l1tx.com

October 27, 2026

Abstract

The LINE-1 retrotransposon is a virus-like genomic element that can activate in diseased cells. It has been implicated in cancer, autoimmunity, developmental disorders, infectious disease and neurodegeneration. LINE-1 can be inhibited by reverse transcriptase inhibitors, small molecules originally developed for HIV. The correlation between HIV and LINE-1 inhibition is ∼0.9 on a log scale, in spite of >450 Mya of evolutionary history since retroviruses diverged from retrotransposons. Suppression of HIV requires prevention of a completed 9.7 kb virus insertion into the genome in immune cells. However, every cell in the body already contains >100 copies of LINE-1 that can potentially activate. Even incomplete LINE-1 mediated insertions can cause damaging mutations and genome instability. Furthermore, partial HIV and LINE-1 products in the cytoplasm of cells can contribute to an inflammatory immune response. Preclinical studies have shown LINE-1 inhibitors slow the progression of otherwise unrelated diseases. LINE-1's contribution to these diseases can potentially be quantified using single-nucleus whole-genome sequencing on diseased tissue, enabling prioritization of disease selection for therapeutic development. Furthermore, LINE-1's ORF1p protein can be measured in the blood of individuals with certain cancers. LINE-1 inhibitors could therefore be extended to various diseases. Potentially used both as an anti-inflammatory and prophylactically to help preserve genome integrity. Reverse transcriptase inhibitors are available at low cost for HIV, and higher doses of some could be used to inhibit LINE-1. However, reverse transcriptase inhibitors do not prevent DNA cutting by LINE-1's endonuclease. Novel inhibitors may therefore be required. This, combined with novel diagnostics opens up possibilities for inhibiting LINE-1 therapeutically in disease

Introduction

One of the biggest impacts on the genome structure and evolution has been retrotransposons. These are virus-like elements that copied and spread over evolutionary time, accounting for ∼43% of the human genome. They also activate under stress in many diseases. This triggers the innate immune system to raise an inflammatory antiviral response, known as 'viral mimicry' (De Cecco et al., 2019). This can contribute to both autoimmunity and immune evasion in cancer (You et al., 2025). They further create genomic instability, which can contribute to both cancer and degeneration (Rodriguez-Martin et al., 2020; Gasior et al., 2006). Retrotransposons are suppressed in healthy cells

The research article below will explain what retrotransposons are and how they can be targeted therapeutically

Evolution of Retrotransposons

Retrotransposons evolved from bacterial group II introns over 1 billion years ago. They are near universal across eukaryotes. However, the naked mole rat, a relatively long-lived species in a highly stable environment with low predation, has lost retrotransposition competent elements (Ivancevic et al., 2016). Although LINE-1 has gone extinct in some other mammalian species (Ivancevic et al., 2016), perhaps through selection or genetic drift, mice contain >2,000 copies of active elements (Goodier et al., 2001). Retroviruses such as HIV evolved from retrotransposon-like precursors, also using reverse transcriptase (RT) to copy their RNA into host genomes (Figure 1A)

LINE-1 in humans

Most retrotransposons in the human genome have been inactivated by mutation, but >100 copies of the LINE-1 retrotransposon are still capable of copying themselves (Yang et al., 2024). LINE-1 copies itself but also other sequences that hijack its machinery, which has resulted in almost half the genome consisting of retrotransposon derived sequences. LINE-1 itself accounts for ∼17% of the human genome in ∼500,000 copies. Active elements are ∼6 kb and encode two proteins, ORF1p and ORF2p (Figure 1A). ORF1p binds to LINE-1 messenger RNA, and ORF2p cuts DNA and reverse transcribes (copies) LINE-1 mRNA back into genomic DNA. This reverse transcription process can occur in the cytoplasm of the cell, producing complementary DNA (Baldwin et al., 2024), which triggers the defense response to viruses initiated by the cell (De Cecco et al., 2019). When DNA is abundant in the cytoplasm this can begin at 45 bp (Andreeva et al., 2017), and at 60 bp for an RNA:DNA hybrid (which is what LINE-1 produces when reverse transcribing) (Mankan et al., 2014). Longer sequences trigger inflammation more efficiently (Du and

Chen, 2018). At the amount of DNA present in an infection, the response can require a few hundred base pairs (Luecke et al., 2017)

There are two types of retrotransposition, classical & non-classical (Figure 1B). Classical retrotransposition results in an insertion with a polyA tail (from the messenger RNA of LINE-1 or hijacked sequence). Insertions often contain target-site duplications (TSD). Non-classical retrotransposition involves no polyA, no TSD, and is truncated at both ends (3′ and 5′). Non-classical is independent of endonuclease (ENi), relying only on reverse transcriptase (Morrish et al., 2002). ENi events have been observed in the brain (Jacob-Hirsch et al., 2018), particularly in certain diseases such as Ataxia-Telangiectasia and Tuberous Sclerosis, but have not (yet) been seen in cancer

LINE-1 can also disrupt the functions of genes and contribute to genome instability. Successful insertions in the sense orientation of genes inside their introns can introduce cryptic splice sites, polyadenylation signals, or otherwise impact gene expression (Perepelitsa-Belancio and Deininger, 2003). Insertions in the population are depleted in introns in a sense orientation relative to genes. Modeled from this, new retrotransposon insertions contribute roughly 5–14% of new loss-of-function gene mutations per birth. Furthermore, insertions inside exons of genes disrupt the gene completely (Cajuso et al., 2019). Failed insertions leave a scar of small insertion/deletion events (Zhang et al., 2026). They can cause more complex rearrangements such as reciprocal translocations between two chromosomes, or inversions of sequence (Zumalave et al., 2026; Mendez-Dorantes et al., 2026). This genome instability can generate further structural rearrangements such as deletions and duplications, complex rearrangements or DNA fragmentation, which is important in cancer (Mendez-Dorantes et al., 2026). Retrotransposons are not the only factor influencing genome integrity. However, the damage they cause can contribute to both apoptosis and cell death as well as to cancer evolution, as we shall see in later sections

How does LINE-1 contribute to disease?

LINE-1 is heavily suppressed in healthy adult cells through regulation of gene expression, sensing and degradation of foreign DNA, autophagy and breakdown of ribonucleoproteins, repair of DNA damage, triggering of cell death or apoptosis (Goodier, 2016). Many different diseases involve dysregulation of genes involved in these processes. LINE-1 is typically not the initiating factor of these diseases but is instead a downstream consequence that can contribute to disease progression. The ORF2p protein of LINE-1 is so toxic to cells that scientists were only recently able to produce enough of it to determine its 3D structure (Leslie, 2025). Genome sequencing has revealed elevated levels of LINE-1 insertions into DNA in both cancer and neurological disorders (Rodriguez-Martin et al., 2020; Jacob-Hirsch et al., 2018)

LINE-1's role in disease

Like many viruses, LINE-1 is both inflammatory and a mutagen. However, everyone contains >100 potentially active copies in every cell in the body. Independent studies have implicated LINE-1 in many diseases (Table 1). These include cancers, autoimmune disorders, developmental disorders, infectious disease and degeneration. The evidence for each disease varies, some have mechanistic explanations, genomic or transcriptomic data, LINE-1 inhibition data, cell line or animal model data, and clinical trial data

LINE-1 inhibitors have slowed progression of otherwise unrelated diseases in preclinical studies, including cancer (Rajurkar et al., 2022; Novototskaya-Vlasova et al., 2022), progeroid syndromes (Simon et al., 2019; Della Valle et al., 2022), Rett syndrome (Martins et al., 2026), Ataxia-Telangiectasia (Takahashi et al., 2022), Aicardi-Goutières syndrome (AGS) (Thomas et al., 2017), Down's syndrome (Martinez de Lagran et al., 2022), Alzheimer's (Herdy et al., 2026), and non-infectious (sterile) inflammation (De Cecco et al., 2019). The degree to which LINE-1 contributes to any of these diseases is not well established. Studies typically use older reverse transcriptase inhibitors such as Lamivudine (3TC) or Stavudine (d4T), because they are readily available at low cost. Both are much less potent than later inhibitors (Figure 2A; Table 2). Stavudine also has a narrow therapeutic window due to mitochondrial polymerase γ toxicity, but the effect nucleosides have on mitochondria varies substantially (Table 2). Studies using less effective nucleosides that may underestimate the potential effects of LINE-1 inhibitors, but they also use continuous high doses that can over-estimate their effects

Although LINE-1 inhibition shows promise, it is not yet resolved what the relative contribution of LINE-1 to each disease. Many studies use indirect methods such as signatures of inflammation or LINE-1 mRNA expression, which are challenging to attribute to elevated retrotransposition accurately (Faulkner, 2023). Whole-genome sequencing has identified elevation of LINE-1 insertions in post-mortem or biopsied brain tissue of individuals with Ataxia-Telangiectasia, Tuberous Sclerosis, and Rett Syndrome (Jacob-Hirsch et al., 2018). It is also controversial how much LINE-1 activation occurs in the brain (Jacob-Hirsch et al., 2018; Upton et al., 2015; Erwin et al., 2016; Evrony et al., 2016). To accurately measure this requires single-nucleus whole-genome sequencing of post-mortem brain tissue. Novel computational biology algorithms are also required,

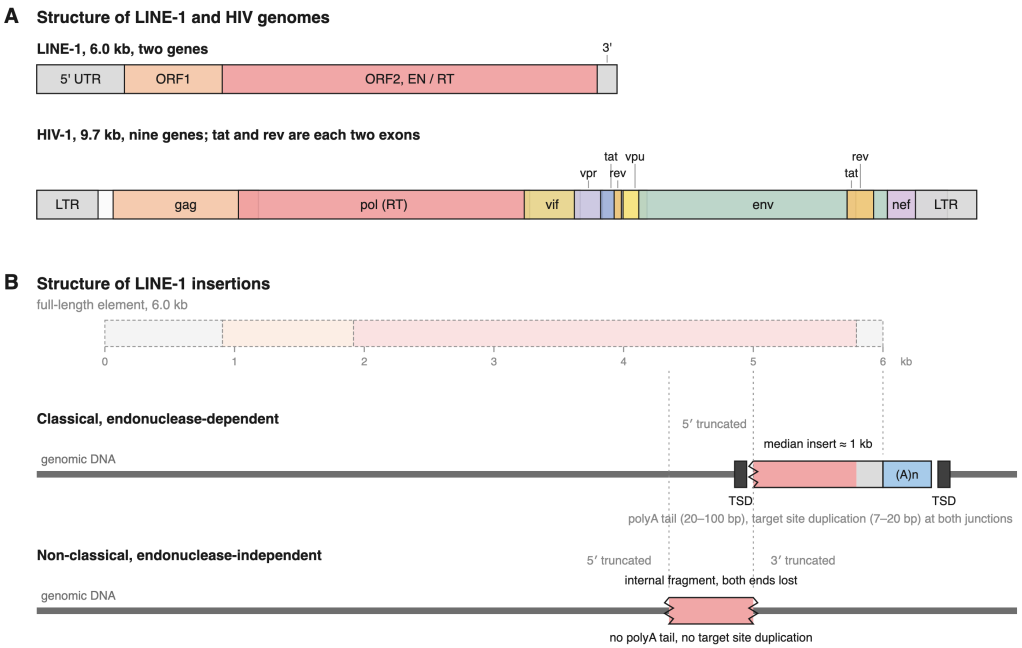


Fig. 1 | Genome structure and insertion classes for LINE-1 and HIV-1 a, LINE-1 is 6.0 kb and encodes ORF1p and ORF2p (endonuclease and reverse transcriptase). HIV-1 is 9.7 kb; reverse transcriptase is in *pol*. **b**, Classical, endonuclease-dependent insertions are usually 5′ truncated and retain a polyA tail and target-site duplications. Non-classical, endonuclease-independent insertions are truncated at both ends

Table 1 | Diseases in which LINE-1 has been implicated

Disease	Category	Gene	LINE-1 mechanism	References
Senescence	Aging	RB1	chromatin relaxation; accumulation of L1 cDNA	[1]
Progeroid syndromes	Aging	LMNA, SIRT6	accumulation of L1 RNA; chromatin relaxation	[21,22]
Ataxia-telangiectasia	Neurodevelopmental	ATM	delayed DNA break repairs	[10,25,37]
Rett syndrome	Neurodevelopmental	MECP2	hypomethylation	[10,23,24]
Down syndrome	Neurodevelopmental		misregulated L1 expression	[27,28]
Tuberous sclerosis	Neurodevelopmental	TSC1, TSC2	mTOR growth signaling activates L1	[10,38]
Cytomegalovirus	Infectious	TSC2	mTOR growth signaling activates L1	[39,40]
Aicardi-Goutières	Inflammatory	TREX1	accumulation of L1 cDNA	[26,36]
Fanconi anemia	Inflammatory	SLX4, FANCD2	accumulation of L1 cDNA	[41]
Sterile inflammation	Inflammatory	CGAS, STING1	accumulation of L1 cDNA	[1]
Lupus	Autoimmune		demethylation; accumulation of L1 RNA	[42]
GI, lung, head & neck, ovarian	Oncological	TP53	demethylation	[3,4,13,14,16,19]
Chemo / radiotherapy	Oncological	TP53	DNA damage / stress induced activation	[19]
ALS / FTD	Neurodegenerative	TARDBP	chromatin relaxation	[43]
Alzheimer / tauopathy	Neurodegenerative	MAPT	chromatin relaxation	[29–31]
Parkinson's	Neurodegenerative		L1 activation in substantia nigra and putamen neurons and microglia	[44,45]
Geographic atrophy	Degenerative	DICER1	L1 reverse transcription of Alu RNA	[46]

such as retrotransposon-miner (`https://github.com/L1-Therapeutics/retrotransposon-miner`)

Another unresolved issue is determining when it is possible to intervene during the course of disease progression. It is possible to measure LINE-1 in real-time in cancer because dying cancer cells release LINE-1 ORF1p protein. However, for neurological disorders or diseases of aging this requires retrospective studies of diseased tissue

Furthermore, LINE-1 is not the only contributor to inflammation. LINE-1 induced cell-death / degeneration cannot

be reversed. Also LINE-1 mutations insert unpredictably and are not the only mutagenic factor in cancer or cell death. LINE-1 inhibitors are not necessarily a cure for any disease, however they may be an important piece of the puzzle in some diseases. They would be used as an anti-inflammatory, or prophylactically to preserve genome stability over a period of years. But the long timelines of disease progression make it both costly and challenging to measure benefits in a traditional clinical trial setting

In spite of promising preclinical studies, there have not yet been approvals for reverse transcriptase inhibitors outside of viral infections. Censavudine has been tested in a phase 2 clinical trial against LINE-1 for amyotrophic lateral sclerosis (ALS) (NCT04993755), Lamivudine for colorectal cancer (NCT03144804; Rajurkar et al., 2022), and a combination of inhibitors for Aicardi-Goutières syndrome (Crow et al., 2025). A phase 1b/2a trial of Lamivudine for Alzheimer's disease in people with Down syndrome was funded by the National Institute on Aging in 2026 and is not yet registered (1UG3AG101657)

Therapeutic Strategies for LINE-1

Three strategies have been proposed for LINE-1, antisense oligonucleotides (ASOs), ORF2p endonuclease inhibitors and ORF2p reverse transcriptase inhibitors (Della Valle et al., 2022; D'Ordine et al., 2024; Jones et al., 2008; Dai et al., 2011). Other approaches include cancer vaccines against ORF1p (McKerrow et al., 2025). However, they will not be discussed here

Reverse transcriptase inhibitors (RTIs)

Reverse transcriptase inhibitors are analogues of nucleosides, the building blocks of DNA. There are nucleoside inhibitors for each nucleoside; adenosine, cytidine, thymidine and guanosine. About 1,000 nucleoside and nucleoside phosphonate analogues have been tested across antiviral programs.[54] For HIV, more than 600 nucleosides have been tested.[53] A further 306 have been tested for LINE-1 activity, and 25 have been tested on both HIV and LINE-1 (Table 2). The potency of RTIs for HIV correlates strongly with their potency for LINE-1 (Figure 2A). Retroviruses diverged evolutionarily from retrotransposons $>$450 Mya (Aiewsakun and Katzourakis, 2017). However, both take deoxyribose nucleoside triphosphates (dNTPs) as substrates, which are unchanged since the dawn of life. Consequently, the shape of the active site of the enzyme has been preserved across evolutionary time (Baldwin et al., 2024). This creates a vulnerability that nucleoside chemists have exploited to develop antivirals. Antivirals work consistently on HIV and LINE-1 reverse transcriptases, and some show substantial selectivity over host enzymes that take dNTPs as substrates (such as mitochondrial polymerase gamma, telomerase, and DNA polymerase). However, LINE-1 may require higher dosing to be sufficiently inhibited, and approved HIV regimens are constructed to provide the minimum necessary dose to sufficiently inhibit HIV. Potency is not the only consideration, as molecules that do inhibit LINE-1 / HIV can also have off-target effects (Table 2), cross the blood-brain barrier with varying efficiency, be activated in only certain cell types, or have different half-lives

Adenosine

Adenosine analogues have specific modifications that prevent their metabolism and increase their selectivity for HIV

Many adenosine analogues have been made, however, we'll just focus on one as an example, Alimatravir, which has a wide therapeutic window and long half-life (2-Chloro-7-deaza-4′-ethynyl-2′-deoxyadenosine; Table 2; Figure 2B; Raheem et al., 2025; Michailidis et al., 2009). It is in Phase 3 at 11 mg monthly for HIV pre-exposure prophylaxis. In peripheral blood cells, that dose is sufficient to keep triphosphate levels above the HIV threshold for one month. No other nucleosides have shown sufficient safety or potency to be dosed only once per month. Alimatravir has no published LINE-1 half-maximal inhibitory concentration (IC_{50}). However, we can estimate it by extrapolating from Figure 2A and scaling by template length. Alimatravir may not be sufficient to block 100-nt or 1000-nt products of LINE-1 throughout monthly dosing (Figure 2C). More frequent dosing may be required, and the safety of this is not established

There are three modifications to the bare nucleoside of Alimatravir that enable its high therapeutic index and long half-life. A 2-chloro group on the adenine base prevents adenosine deaminase from metabolizing the nucleoside to inosine (Raheem et al., 2025). The 4′ ethynyl group on the ribose sugar blocks SAMHD1 from breaking down the triphosphate, as well as incorporating more selectively into HIV/LINE-1 (Michailidis et al., 2009). Finally, the 7-deaza group prevents breakdown by purine nucleoside phosphorylase and also incorporation into host polymerases (Figure 2B; Raheem et al., 2025)

Adenine has two positions that can be altered while sparing Watson-Crick base pairing, whereas cytosine and thymine are mostly limited to one position. Both of those positions are also used by catabolic enzymes to break down the nucleoside. Modifying them enables both a longer half life and selectivity for HIV / LINE-1

Although there is no public measurement of LINE-1 inhibition for Alimatravir, nor data on its ability to cross the blood brain barrier. We can assuming the data will be similar to Islatravir (EFdA; Figure 2A), another adenosine

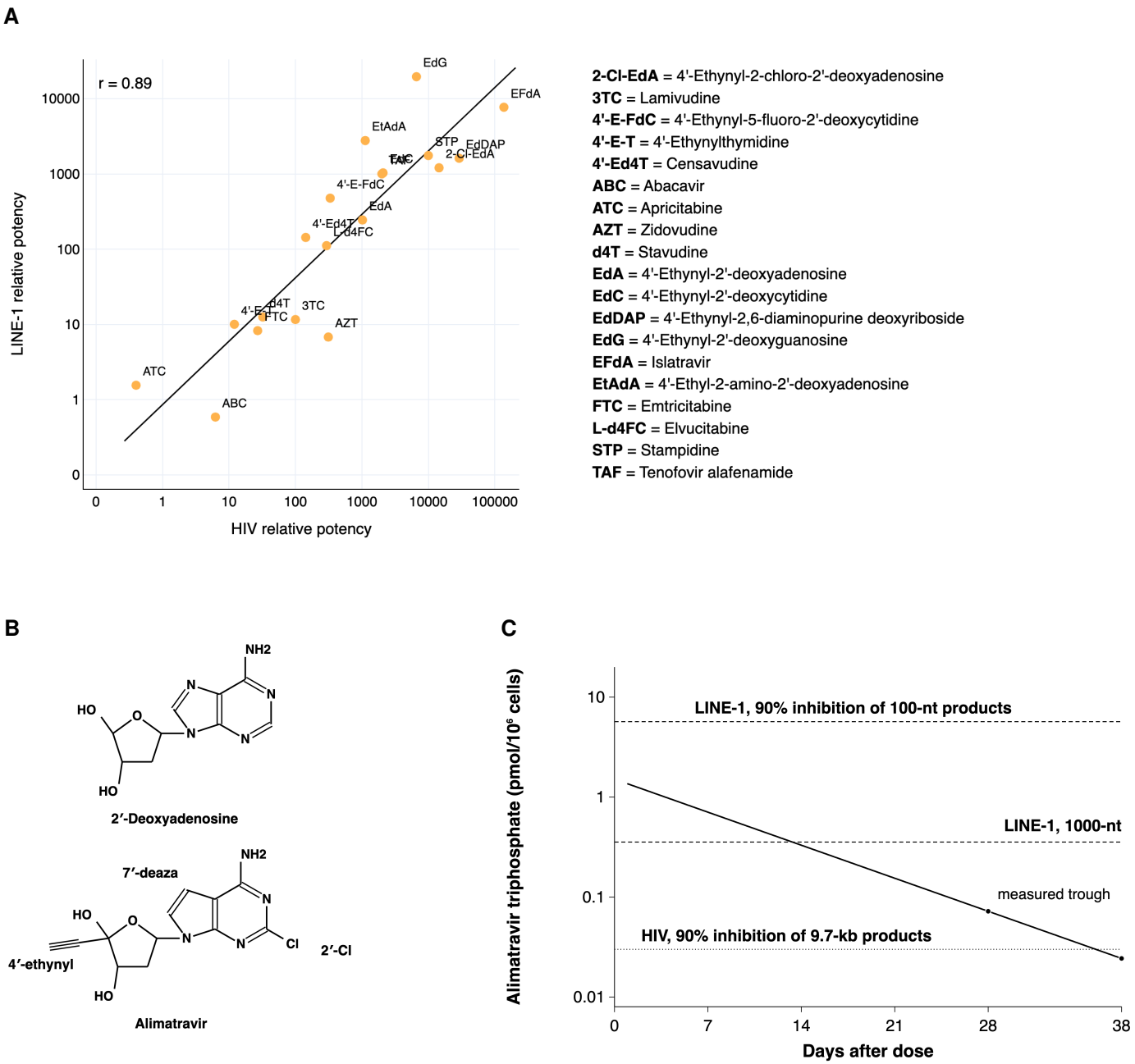


Fig. 2 | Nucleoside potency, structure, and exposure. a, Relative potency of nucleoside analogues. LINE-1 potency tracks HIV potency. **b**, Deoxyadenosine and alimatravir. Alimatravir is an example reverse transcriptase inhibitor, 2-chloro-7-deaza-4′-ethynyl-2′-deoxyadenosine. **c**, Intracellular alimatravir triphosphate after a monthly 11 mg oral dose, in pmol per 10^6 peripheral blood mononuclear cells, on a logarithmic scale. The upper line is the calculated concentration at which 90% of 100-nt products would be inhibited. The middle line is the same calculation for 1,000-nt products. The LINE-1 IC_{50} is extrapolated from panel A, then scaled by template length. The lower line is the published threshold for 90% inhibition of a 9.7 kb HIV product. Data are from US20240173346A1, US20230293526A1, US20240366649A1, US20260035403A1, US12187758B2, EP1177202B1[52,68,69,71–73,76–78]

analogue from the same series as Alimatravir (Stoddart et al., 2015)

Thymidine

When LINE-1 is reverse transcribed it starts copying the polyA tail of its messenger RNA, which is ~40 nucleotides long and is complementary to thymidine (Figure 1). Thymidine analogues therefore could in theory shut down LINE-1 reverse transcription before it is able to contribute to inflammation, Thymidine is complementary to polyA sequence. Thymidine nucleosides primarily get activated by thymidine kinase 1 (TK1). This enzyme is only present in dividing cells, as thymidine is required only for DNA (not for RNA). Whereas resting cells can synthesize dTTP as needed, rendering thymidine analogues potentially less effective in

Table 2 | Nucleoside analogues tested on LINE-1 and HIV. Concentrations are nM

Compound	Short	Base	LINE-1 IC_{50}	HIV EC_{50}	CC_{50}	Pol γ	Pol α	Telomerase
Alimatravir	MK-8527	A		0.21	>42,000	>200,000	95,000	
Islatravir	EFdA	A	1.3	0.073	7,500	10,000	>100,000	
4′-Ethynyl-2′-deoxyguanosine	EdG	G	0.51	1.51	1,400			
4′-Ethynyl-2,6-diaminopurine deoxyriboside	EdDAP	A	6.2	0.344	900			
4′-Ethyl-2-amino-2′-deoxyadenosine	EtAdA	A	3.6	8.9	35,900			
4′-Ethynyl-2′-deoxycytidine	EdC	C	9.7	4.8	2,200			
Tenofovir alafenamide	TAF	A	10	5	4,700			13,000
ROME compound I-1	I-1	A	8.3		>100,000			
4′-vinyl-5-fluoro-2′-deoxycytidine	ROME-74	C	9		>100,000			
4′-vinyl-2′-deoxyguanosine	ROME-69	G	13		>1,000–≤10,000			
4′-Ethynyl-5-fluoro-2′-deoxycytidine	4′-E-FdC	C	21	30	>100,000			
4′-chloromethyl-2′-deoxycytidine	ROME-2	C	<50		>100,000	≤50,000		≤1,000
2′-fluoro-4′-chloromethyl-5-fluoro-2′-deoxycytidine	ROME-67	C	<50		>100,000	>200,000–≤300,000		>100,000
Elvucitabine	L-d4FC	C	90	34				
Censavudine	4′-Ed4T	T	70	70	>100,000	100,000	>100,000	
Lamivudine	3TC	C	640	100	>100,000			>150,000
Zidovudine	AZT	T	810	32	29,000			46,000
Stavudine	d4T	T	750	310	79,000	1,000		15,000
Emtricitabine	FTC	C	1,340	370				>150,000
4′-Ethynylthymidine	4′-E-T	T	>1,000	830	>400,000			
Abacavir	ABC	G	17,100	1,600				9,000

Data are from US20240173346A1, US20230293526A1, US20240366649A1, US20260035403A1, US12187758B2, EP1177202B1[8,52,64,65,68–77]

non-dividing cells such as neurons (Sherley and Kelly, 1988). Thymidine analogues therefore may be particularly beneficial in dividing cells such as in cancer, especially given the chemotherapy standard of care (SoC) leverages thymidine synthesis inhibitors (5-FU), which is rationally synergistic with thymidine analogues

Mitochondria contain their own thymidine kinase, which can activate nucleosides and contribute to mitochondrial toxicity (Arnér and Eriksson, 1995). The thymidine analogue censavudine has a 4′ethynyl modification (like Alimatravir), which enables both a strong LINE-1 potency and orders of magnitude lower incorporation into mitochondrial polymerase gamma (Yang et al., 2007)

Unfortunately, the cell efficiently turns over thymidine-triphosphates. Thymidine is also cleared by thymidine phosphorylase cutting its glycosidic bond. Although Censavudine's 4′ethynyl resists this process, its half life is still much shorter than that of adenosine analogues such as Alimatravir. Censavudine has been used in a Phase 2 trial for amyotrophic lateral sclerosis / frontotemporal dementia (NCT04993755). The ALS functional rating (ALSFRS-R) did not differ from placebo over the 24-week blinded period, which may in part be due to lack of activation in neurons

Cytidine

Two of the most commonly prescribed nucleosides for HIV are cytidine analogues, Lamivudine (3TC) and Emtricitabine (FTC). Although their potency is not as high as for other molecules, they have strong safety records due to having opposite chirality to natural nucleosides, alongside other modifications (Coates et al., 1992). Lamivudine has been tested in a Phase 2 trial for stage IV colorectal cancer (NCT03144804), and showed some suggestion of disease stabilization, but had no control arm (Rajurkar et al., 2022). Lamivudine at standard doses is not potent enough to sufficiently suppress LINE-1. ROME therapeutics screened cytidine nucleosides for LINE-1, generating novel compounds that have greater potency than 3TC or FTC, and which may be in the same range as the adenosine analogues (ROME Therapeutics, US12187758B2; Table 2). However, their safety is not established. It is not known if any are as potent as Alimatravir, as published data had a limit of detection of 50 nM (Table 2). One advantage cytidine analogues has is they are not incorporated into telomerase because the enzyme adds a fixed TTAGGG repeat (Table 2) and does not use cytidine

Guanosine

Guanosine analogues have historically had issues with toxicity. The most potent known nucleoside for LINE-1 is 4′-Ethynyl-2′-deoxyguanosine (0.51 nM) (Table 2; Figure 2A). In six-week-old male mice, one dose of 4′-ethynyl-deoxyguanosine also killed every animal at 3 mg/kg within a few days (Ohrui, 2011), but 1mg / kg did not. In the same experiment, 4′-ethynyl-deoxyadenosine was tolerated at 100 mg/kg. Although highly potent (Figure 2A), none are approved for HIV. Deoxyguanosine analogues are phosphorylated mainly by mitochondrial deoxyguanosine kinase, which may contribute to toxicity due to proximity to mitochondrial polymerase gamma. Guanosine analogues

are underexplored, and like adenosine have two positions on the base that can be modified without impacting Watson-Crick base pairing. However, LINE-1 is AT rich (∼57%) and adenosine and thymidine analogues apperar preferable

Prodrug formulations

Bare nucleosides of guanosine, cytidine and adenosine cannot easily be combined because they compete with each other for phosphorylation by cellular kinase enzymes (Arnér and Eriksson, 1995). In theory this is less problematic for the potent nucleosides, because the ratio of nucleoside to natural dNTPs is lower. Tenofovir alafenamide (TAF) is a prodrug formulation that contains a phosphonate group that has been masked, enabling combinations of the nucleosides as the first phosphorylation step is not necessary to activate TAF. While potent for HIV and LINE-1 (Callebaut et al., 2015; Table 2), the downside of this approach is the prodrug formulations do not typically cross the blood-brain barrier as well as bare nucleosides (Best et al., 2012). Also adenosines are readily phosphorylated, and both islatravir & alimatravir are more potent than TAF while being bare nucleosides (Table 2)

ORF1p / ORF2p inhibitors

Even if LINE-1 reverse transcription is completely shut down, this does not prevent LINE-1's ORF2p endonuclease domain from cutting DNA. This is so toxic to cells that scientists were not able to produce enough of the protein to determine its structure until 2023 (Baldwin et al., 2024; Leslie, 2025; Thawani et al., 2024). One attempt has been made to make a non-nucleoside inhibitor that inhibits the endonuclease domain of ORF2p (D'Ordine et al., 2024). One challenge is that the endonuclease shares conserved catalytic sites with host DNA repair enzymes such as APE1. Just like with HIV, a combination of a reverse transcriptase inhibitor with a non-nucleoside small molecule inhibitor may be necessary to shut down LINE-1's activity

Antisense Oligonucleotides

Antisense oligonucleotides (ASOs) could potentially prevent the translation of LINE-1 mRNA into ORF1p & ORF2p. They have shown promise for LINE-1 in preclinical models of Hutchinson-Gilford progeria (Della Valle et al., 2022) and Alzheimer's (Herdy et al., 2026). However, LINE-1 is 17% of the genome. All ASO designs for LINE-1 have hundreds of potential off-target effects in introns of pre-mRNA. There are two types of ASO, a gapmer, which is modified RNA nucleotides with a DNA gap that recruits RNase H to cut the bound RNA catalytically, and steric blockers, which occupy the target site, preventing translation. For gapmers, LINE-1 ASOs would also degrade potentially important coding gene transcripts. For steric blockers their effects would be diluted by off-target binding, requiring high doses. Short interfering RNAs (siRNAs) also present similar challenges of off-target effects

Detecting LINE-1 activity in disease

Having effective inhibitors against LINE-1 is not sufficient. We also need to know under what contexts to administer them. To date more progress has been made in detecting LINE-1 activity in cancer than in other diseases. Tumor cells are clonal, enabling detection of retrotransposons using bulk whole-genome sequencing (Rodriguez-Martin et al., 2020; Cajuso et al., 2019; Zhang et al., 2026; Zumalave et al., 2026; Mendez-Dorantes et al., 2026). Furthermore, dying cancer cells shed proteins such as ORF1p into blood, enabling detection of LINE-1 activity from a blood draw (Taylor et al., 2023)

It is now possible to detect retrotransposons jumping in individual cells through single-nucleus whole-genome sequencing. Although not yet demonstrated for LINE-1, structural variants can be detected in single nuclei, and are detected using similar signatures (Luquette et al., 2025). Bioinformatics tools such as retrotransposon-miner could be adapted to single-nucleus sequencing for this purpose (). Retrotransposon-miner is developed by L1 Therapeutics (`https://l1tx.com`). This opens the possibility of sequencing single nuclei from both post-mortem tissue and biopsies to determine what diseases and what time points in a disease are most amenable to LINE-1 inhibitors

Discussion and Future Perspectives

The challenge with all diseases where LINE-1 is implicated is that treatment with inhibitors is not likely to be a cure for that disease, instead they may slow or stop progression. For example, in cancer LINE-1 causes mutations that can contribute to tumor evolution. But it is not the only source of mutations, and although preclinical studies have shown LINE-1 inhibition can slow tumor cell growth or division, it is not understood how well that would translate into the clinic. Similar undertainties remain for degeneration and inflammation. Most studies in these diseases are preliminary, and we do not have a clear picture of the relative importance of LINE-1 to each disease. New technologies such as single-nucleus whole-genome sequencing enable the quantification of the prevalence of insertions in diseased tissue across a range of ages and disease stages. This would help measure LINE-1's contribution to genome instability, and act as a proxy for its inflammatory effects. Measuring this requires development of novel computational biology algorithms

The potency, safety and low cost of next generation reverse transcriptase inhibitors potentially enable their application to LINE-1. However, this also creates a challenge

for expanding access for nucleosides to LINE-1, because the cost of performing the necessary preclinical work and trials to determine safe dosing and when treatment would be effective may exceed what can be recouped due to pricing pressure from off-label prescribing

However, reverse transcriptase inhibitors alone may not be sufficient for LINE-1, even at higher doses. One concern that has been raised about them is that stopping reverse transcriptase early can free up ORF2p endonuclease to repeatedly cut DNA (Rajurkar et al., 2022). However, this is unlikely given ORF2p is unstable without RNA bound (Baldwin et al., 2024), and reverse transcriptase inhibitors reduce DNA damage rather than increasing it (D'Ordine et al., 2024). Also, the success of reverse transcriptase inhibitors outside of cancer in preclinical studies argues against this hypothesis

Regardless, a combination of inhibitors may be required for LINE-1 because reverse transcriptase inhibitors only partially prevent DNA damage (D'Ordine et al., 2024). More than 10 DNA cuts are made for each successful insertion (Gasior et al., 2006). A selective inhibitor that blocks endonuclease activity or degrades the protein could provide greater protection against LINE-1 activation. As of October 2026, there is no publicly announced program aimed at the ORF2p endonuclease

LINE-1 is an unconventional target for new medicines. It encodes proteins, but is not a gene. It is virus-like, but not infectious. It is suppressed in healthy adults and its activation has been associated with many diseases. Evidence increasingly suggests therapeutics for LINE-1 could have broad reach, however we have not established precisely when diseases are therapeutically actionable. This creates both challenges and opportunities for inhibiting LINE-1 therapeutically in disease

Competing interests

W.B. is founder of L1 Therapeutics. The retrotransposon-miner software cited in this article is developed by L1 Therapeutics

References

1. De Cecco, M. et al. LINE-1 derepression in senescent cells triggers interferon and inflammaging. *Nature* **566**, 73–78 (2019) doi:10.1038/s41586-018-0784-9

2. You, E. et al. LINE-1 ORF1p mimics viral innate immune evasion mechanisms in pancreatic ductal adenocarcinoma. *Cancer Discov.* **15**, 1063–1082 (2025) doi:10.1158/2159-8290.cd-24-1317

3. Rodriguez-Martin, B. et al. Pan-cancer analysis of whole genomes identifies driver rearrangements promoted by LINE-1 retrotransposition. *Nat. Genet.* **52**, 306–319 (2020) doi:10.1038/s41588-019-0562-0

4. Gasior, S. L., Wakeman, T. P., Xu, B. & Deininger, P. L. The human LINE-1 retrotransposon creates DNA double-strand breaks. *J. Mol. Biol.* **357**, 1383–1393 (2006) doi:10.1016/j.jmb.2006.01.089

5. Ivancevic, A. M., Kortschak, R. D., Bertozzi, T. & Adelson, D. L. LINEs between Species: Evolutionary Dynamics of LINE-1 Retrotransposons across the Eukaryotic Tree of Life. *Genome Biol. Evol.* **8**, 3301–3322 (2016) doi:10.1093/gbe/evw243

6. Goodier, J. L., Ostertag, E. M., Du, K. & Kazazian, H. H. A novel active L1 retrotransposon subfamily in the mouse. *Genome Res.* **11**, 1677–1685 (2001) doi:10.1101/gr.198301

7. Yang, L. et al. Evolutionary insights from profiling LINE-1 activity at allelic resolution in a single human genome. *EMBO J.* **43**, 112–131 (2024) doi:10.1038/s44318-023-00007-y

8. Baldwin, E. T. et al. Structures, functions and adaptations of the human LINE-1 ORF2 protein. *Nature* **626**, 194–206 (2024) doi:10.1038/s41586-023-06947-z

9. Morrish, T. A. et al. DNA repair mediated by endonuclease-independent LINE-1 retrotransposition. *Nat. Genet.* **31**, 159–165 (2002) doi:10.1038/ng898

10. Jacob-Hirsch, J. et al. Whole-genome sequencing reveals principles of brain retrotransposition in neurodevelopmental disorders. *Cell Res.* **28**, 187–203 (2018) doi:10.1038/cr.2018.8

11. Perepelitsa-Belancio, V. & Deininger, P. RNA truncation by premature polyadenylation attenuates human mobile element activity. *Nat. Genet.* **35**, 363–366 (2003) doi:10.1038/ng1269

12. Cajuso, T. et al. Retrotransposon insertions can initiate colorectal cancer and are associated with poor survival. *Nat. Commun.* **10**, 4022 (2019) doi:10.1038/s41467-019-11770-0

13. Zhang, T. et al. Uncovering the role of LINE-1 in the evolution of lung adenocarcinoma. *Nature* **650**, 230–241 (2026) doi:10.1038/s41586-025-09825-y

14. Pisanic, T. R. et al. Long interspersed nuclear element 1 retrotransposons become deregulated during the development of ovarian cancer precursor lesions. *Am. J. Pathol.* **189**, 513–520 (2019) doi:10.1016/j.ajpath.2018.11.005

15. Zumalave, S. et al. Concurrent L1 retrotransposition events promote reciprocal translocations in human tumorigenesis. *Science* **392**, eaee4513 (2026) doi:10.1126/science.aee4513

16. Mendez-Dorantes, C. et al. Chromosomal rearrangements and instability caused by LINE-1 retrotransposition. 2024.12.14.628481 Preprint at `https://doi.org/10.1101/2024.12.14.628481` (2026)

17. Goodier, J. L. Restricting retrotransposons: a review. *Mob. DNA* **7**, 16 (2016) doi:10.1186/s13100-016-0070-z

18. Leslie, M. Risky moves. *Science* **388**, 244–247 (2025) doi:10.1126/science.ady2285

19. Rajurkar, M. et al. Reverse Transcriptase Inhibition Disrupts Repeat Element Life Cycle in Colorectal Cancer. *Cancer Discov.* **12**, 1462–1481 (2022) doi:10.1158/2159-8290.cd-21-1117

20. Novototskaya-Vlasova, K. A. et al. Inflammatory response to retrotransposons drives tumor drug resistance that can be prevented by reverse transcriptase inhibitors. *Proc. Natl. Acad. Sci. U. S. A.* **119**, e2213146119 (2022) doi:10.1073/pnas.2213146119

21. Simon, M. et al. LINE1 Derepression in Aged Wild-Type and SIRT6-Deficient Mice Drives Inflammation. *Cell Metab.* **29**, 871–885.e5 (2019) doi:10.1016/j.cmet.2019.02.014

22. Della Valle, F. et al. LINE-1 RNA causes heterochromatin erosion and is a target for amelioration of senescent phenotypes in progeroid syndromes. *Sci. Transl. Med.* **14**, eabl6057 (2022) doi:10.1126/scitranslmed.abl6057

23. Muotri, A. R. et al. L1 retrotransposition in neurons is modulated by MeCP2. *Nature* **468**, 443–446 (2010) doi:10.1038/nature09544

24. Martins, A. M. A. et al. Dormant viral pathways underlie space-induced neural senescence. 2025.11.02.686043 Preprint at `https://doi.org/10.1101/2025.11.02.686043` (2026)

25. Takahashi, T. et al. LINE-1 activation in the cerebellum drives ataxia. *Neuron* **110**, 3278–3287.e8 (2022) doi:10.1016/j.neuron.2022.08.011

26. Thomas, C. A. et al. Modeling of TREX1-Dependent Autoimmune Disease using Human Stem Cells Highlights L1 Accumulation as a Source of Neuroinflammation. *Cell Stem Cell* **21**, 319–331.e8 (2017) doi:10.1016/j.stem.2017.07.009

27. Martinez de Lagran, M., Elizalde-Torrent, A., Paredes, R., Clotet, B. & Dierssen, M. Lamivudine, a reverse transcriptase inhibitor, rescues cognitive deficits in a mouse model of down syndrome. *J. Cell. Mol. Med.* **26**, 4210–4215 (2022) doi:10.1111/jcmm.17411

28. Borgognone, A. et al. Lamivudine modulates the expression of neurological impairment-related genes and LINE-1 retrotransposons in brain tissues of a Down syndrome mouse model. *Front. Aging Neurosci.* **16**, 1386944 (2024) doi:10.3389/fnagi.2024.1386944

29. Guo, C. et al. Tau activates transposable elements in Alzheimer's disease. *Cell Rep.* **23**, 2874–2880 (2018) doi:10.1016/j.celrep.2018.05.004

30. Sun, W. et al. Pathogenic tau-induced piRNA depletion promotes neuronal death through transposable element dysregulation in neurodegenerative tauopathies. *Nat. Neurosci.* **21**, 1038–1048 (2018) doi:10.1038/s41593-018-0194-1

31. Herdy, J. R. et al. Cytoplasmic DNA Sensing Links LINE-1 Expression to Neuronal Senescence in Alzheimer's Disease. 2026.07.27.740588 Preprint at `https://doi.org/10.64898/2026.07.27.740588` (2026)

32. Faulkner, G. J. Elevated L1 expression in ataxia telangiectasia likely explained by an RNA-seq batch effect. *Neuron* **111**, 610–611 (2023) doi:10.1016/j.neuron.2023.02.007

33. Upton, K. R. et al. Ubiquitous L1 mosaicism in hippocampal neurons. *Cell* **161**, 228–239 (2015) doi:10.1016/j.cell.2015.03.026

34. Erwin, J. A. et al. L1-associated genomic regions are deleted in somatic cells of the healthy human brain. *Nat. Neurosci.* **19**, 1583–1591 (2016) doi:10.1038/nn.4388

35. Evrony, G. D., Lee, E., Park, P. J. & Walsh, C. A. Resolving rates of mutation in the brain using single-neuron genomics. *eLife* **5**, e12966 (2016) doi:10.7554/elife.12966

36. Crow, Y. J. et al. Reverse transcriptase inhibitors in Aicardi–Goutières syndrome: A crossover clinical trial. *Dev. Med. Child Neurol.* **67**, 750–757 (2025) doi:10.1111/dmcn.16199

37. Coufal, N. G. et al. Ataxia telangiectasia mutated (ATM) modulates long interspersed element-1 (L1) retrotransposition in human neural stem cells. *Proc. Natl. Acad. Sci.* **108**, 20382–20387 (2011) doi:10.1073/pnas.1100273108

38. Martin, K. R. et al. The genomic landscape of tuberous sclerosis complex. *Nat. Commun.* **8**, 15816 (2017) doi:10.1038/ncomms15816

39. Hwang, S.-Y. et al. Human cytomegalovirus harnesses host L1 retrotransposon for efficient replication. *Nat. Commun.* **15**, 7640 (2024) doi:10.1038/s41467-024-51961-y

40. Park, S., Jeong, J. & Ahn, K. Human cytomegalovirus infection induces L1 expression through UL38-dependent mTOR-KAP1 pathway. *PLoS ONE* **20**, e0320512 (2025) doi:10.1371/journal.pone.0320512

41. Brégnard, C. et al. Upregulated LINE-1 Activity in the Fanconi Anemia Cancer Susceptibility Syndrome Leads to Spontaneous Pro-inflammatory Cytokine Production. *eBioMedicine* **8**, 184–194 (2016) doi:10.1016/j.ebiom.2016.05.005

42. Mavragani, C. P. et al. Expression of Long Interspersed Nuclear Element 1 Retroelements and Induction of Type I Interferon in Patients With Systemic Autoimmune Disease. *Arthritis Rheumatol.* **68**, 2686–2696 (2016) doi:10.1002/art.39795

43. Liu, E. Y. et al. Loss of Nuclear TDP-43 Is Associated with Decondensation of LINE Retrotransposons. *Cell Rep.* **27**, 1409–1421.e6 (2019) doi:10.1016/j.celrep.2019.04.003

44. Garza, R. et al. Activation of transposable elements is linked to a region- and cell-type-specific interferon response in Parkinson's disease. 2025.09.03.673956 Preprint at `https://doi.org/10.1101/2025.09.03.673956` (2026)

45. Gordillo-González, F. et al. A single-cell meta-analysis evidences transposable element dysregulation in sex-based differences in Parkinson's disease. *Neurobiol. Dis.* **229**, 107580 (2026) doi:10.1016/j.nbd.2026.107580

46. Fukuda, S. et al. Alu complementary DNA is enriched in atrophic macular degeneration and triggers retinal pigmented epithelium toxicity via cytosolic innate immunity. *Sci. Adv.* **7**, eabj3658 (2021) doi:10.1126/sciadv.abj3658

47. D'Ordine, A. M., Jogl, G. & Sedivy, J. M. Identification and characterization of small molecule inhibitors of the LINE-1 retrotransposon endonuclease. *Nat. Commun.* **15**, 3883 (2024) doi:10.1038/s41467-024-48066-x

48. Jones, R. B. et al. Nucleoside Analogue Reverse Transcriptase Inhibitors Differentially Inhibit Human LINE-1 Retrotransposition. *PLOS ONE* **3**, e1547 (2008) doi:10.1371/journal.pone.0001547

49. Dai, L., Huang, Q. & Boeke, J. D. Effect of reverse transcriptase inhibitors on LINE-1 and Ty1 reverse transcriptase activities and on LINE-1 retrotransposition. *BMC Biochem.* **12**, 18 (2011) doi:10.1186/1471-2091-12-18

50. McKerrow, W. et al. LINE-1 ORF1p is a shared and immunogenic antigen in cancer. 2025.06.11.659092 Preprint at `https://doi.org/10.1101/2025.06.11.659092` (2025)

51. Aiewsakun, P. & Katzourakis, A. Marine origin of retroviruses in the early Palaeozoic Era. *Nat. Commun.* **8**, 13954 (2017) doi:10.1038/ncomms13954

52. Raheem, I. T. et al. MK-8527 is a novel inhibitor of HIV-1 reverse transcriptase translocation with potential for extended-duration dosing. *PLOS Biol.* **23**, e3003308 (2025) doi:10.1371/journal.pbio.3003308

53. Raheem, I. T. et al. Discovery of MK-8527, a long-acting HIV nucleoside reverse transcriptase translocation inhibitor. Conference on Retroviruses and Opportunistic Infections, poster 638 (2024). https://www.croiconference.org/wp-content/uploads/sites/2/posters/2024/638.pdf

54. Siegel, D. et al. Discovery and synthesis of a phosphoramidate prodrug of a pyrrolo[2,1-f][1,2,4]triazin-4-amino adenine C-nucleoside (GS-5734) for the treatment of Ebola and emerging viruses. *J. Med. Chem.* **60**, 1648–1661 (2017) doi:10.1021/acs.jmedchem.6b01594

55. Michailidis, E. et al. Mechanism of inhibition of HIV-1 reverse transcriptase by 4′-Ethynyl-2-fluoro-2′-deoxyadenosine triphosphate, a translocation-defective reverse transcriptase inhibitor. *J. Biol. Chem.* **284**, 35681–35691 (2009) doi:10.1074/jbc.m109.036616

56. Stoddart, C. A. et al. Oral administration of the nucleoside EFdA (4′-ethynyl-2-fluoro-2′-deoxyadenosine) provides rapid suppression of HIV viremia in humanized mice and favorable pharmacokinetic properties in mice and the rhesus macaque. *Antimicrob. Agents Chemother.* **59**, 4190–4198 (2015) doi:10.1128/aac.05036-14

57. Andreeva, L. et al. cGAS senses long and HMGB/TFAM-bound U-turn DNA by forming protein–DNA ladders. *Nature* **549**, 394–398 (2017) doi:10.1038/nature23890

58. Du, M. & Chen, Z. J. DNA-induced liquid phase condensation of cGAS activates innate immune signaling. *Science* **361**, 704–709 (2018) doi:10.1126/science.aat1022

59. Luecke, S. et al. cGAS is activated by DNA in a length-dependent manner. *EMBO Rep.* **18**, 1707–1715 (2017) doi:10.15252/embr.201744017

60. Mankan, A. K. et al. Cytosolic RNA:DNA hybrids activate the cGAS–STING axis. *EMBO J.* **33**, 2937–2946 (2014) doi:10.15252/embj.201488726

61. Sherley, J. L. & Kelly, T. J. Regulation of human thymidine kinase during the cell cycle. *J. Biol. Chem.* **263**, 8350–8358 (1988) doi:10.1016/s0021-9258(18)68484-4

62. Arnér, E. S. J. & Eriksson, S. Mammalian deoxyribonucleoside kinases. *Pharmacol. Ther.* **67**, 155–186 (1995) doi:10.1016/0163-7258(95)00015-9

63. Coates, J. A. et al. (−)-2′-deoxy-3′-thiacytidine is a potent, highly selective inhibitor of human immunodeficiency virus type 1 and type 2 replication in vitro. *Antimicrob. Agents Chemother.* **36**, 733–739 (1992) doi:10.1128/aac.36.4.733

64. Ohrui, H. Development of modified nucleosides that have supremely high anti-HIV activity and low toxicity and prevent the emergence of resistant HIV mutants. *Proc. Jpn. Acad. Ser. B Phys. Biol. Sci.* **87**, 53–65 (2011) doi:10.2183/pjab.87.53

65. Callebaut, C., Stepan, G., Tian, Y. & Miller, M. D. In Vitro Virology Profile of Tenofovir Alafenamide, a Novel Oral Prodrug of Tenofovir with Improved Antiviral Activity Compared to That of Tenofovir Disoproxil Fumarate. *Antimicrob. Agents Chemother.* **59**, 5909–5916 (2015) doi:10.1128/aac.01152-15

66. Best, B. M. et al. Low CSF Concentrations of the Nucleotide HIV Reverse Transcriptase Inhibitor, Tenofovir. *J. Acquir. Immune Defic. Syndr. 1999* **59**, 376–381 (2012) doi:10.1097/qai.0b013e318247ec54

67. Thawani, A., Florez Ariza, A. J., Nogales, E. & Collins, K. Template and target-site recognition by human LINE-1 in retrotransposition. *Nature* **626**, 186–193 (2024) doi:10.1038/s41586-023-06933-5

68. Kawamoto, A. et al. 2′-Deoxy-4′-C-ethynyl-2-halo-adenosines active against drug-resistant human immunodeficiency virus type 1 variants. *Int. J. Biochem. Cell Biol.* **40**, 2410–2420 (2008) doi:10.1016/j.biocel.2008.04.007

69. Kodama, E. N. et al. 4′-Ethynyl nucleoside analogs: potent inhibitors of multidrug-resistant human immunodeficiency virus variants in vitro. *Antimicrob. Agents Chemother.* **45**, 1539–1546 (2001) doi:10.1128/aac.45.5.1539-1546.2001

70. Nakata, H. et al. Activity against human immunodeficiency virus type 1, intracellular metabolism, and effects on human DNA polymerases of 4′-ethynyl-2-fluoro-2′-deoxyadenosine. *Antimicrob. Agents Chemother.* **51**, 2701–2708 (2007) doi:10.1128/aac.00277-07

71. Takamatsu, Y. et al. The high genetic barrier of EFdA/MK-8591 stems from strong interactions with the active site of drug-resistant HIV-1 reverse transcriptase. *Cell Chem. Biol.* **25**, 1268–1278.e3 (2018) doi:10.1016/j.chembiol.2018.07.014

72. Lin, T.-S. et al. Design and synthesis of 2′,3′-dideoxy-2′,3′-didehydro-β-L-cytidine and 2′,3′-dideoxy-2′,3′-didehydro-β-L-5-fluorocytidine, potent inhibitors of human immunodeficiency virus in vitro. *J. Med. Chem.* **39**, 1757–1759 (1996) doi:10.1021/jm950836q

73. Nitanda, T. et al. Anti-human immunodeficiency virus type 1 activity and resistance profile of 2′,3′-didehydro-3′-deoxy-4′-ethynylthymidine in vitro. *Antimicrob. Agents Chemother.* **49**, 3355–3360 (2005) doi:10.1128/aac.49.8.3355-3360.2005

74. Haraguchi, K. et al. From the chemistry of epoxy-sugar nucleosides to the discovery of anti-HIV agent 4′-ethynylstavudine-Festinavir. *Curr. Pharm. Des.* **19**, 1880–1897 (2013) doi:10.2174/1381612811319100011

75. Yang, G. et al. Highly selective action of triphosphate metabolite of 4′-ethynyl D4T: a novel anti-HIV compound against HIV-1 RT. *Antiviral Res.* **73**, 185–191 (2007) doi:10.1016/j.antiviral.2006.10.002

76. Hachiya, A. et al. Evaluation of combinations of 4′-ethynyl-2-fluoro-2′-deoxyadenosine with clinically used antiretroviral drugs. *Antimicrob. Agents Chemother.* **57**, 4554–4558 (2013) doi:10.1128/aac.00283-13

77. Parkin, N. T. et al. Natural variation of drug susceptibility in wild-type human immunodeficiency virus type 1. *Antimicrob. Agents Chemother.* **48**, 437–443 (2004) doi:10.1128/aac.48.2.437-443.2004

78. Mayer, K. H. et al. Alimatravir oral once monthly in adults with low likelihood of HIV exposure: a randomized, double-blind, placebo-controlled study. *J. Infect. Dis.* (2026) doi:10.1093/infdis/jiag439

79. Taylor, M. S. et al. Ultrasensitive detection of circulating LINE-1 ORF1p as a specific multicancer biomarker. *Cancer Discov.* **13**, 2532–2547 (2023) doi:10.1158/2159-8290.cd-23-0313

80. Luquette, L. J. et al. A comprehensive view of somatic mosaicism by single-cell DNA analysis. 2025.10.31.685648 Preprint at https://doi.org/10.1101/2025.10.31.685648 (2025)